

MedCite Installation Guide

This guide is for judges/reviewers to run or verify the project quickly.

1) Tech Stack

- Frontend: Next.js 16 (App Router, JavaScript) + Tailwind CSS 4 + shadcn/ui
- Backend: FastAPI + LanceDB + sentence-transformers (all-MiniLM-L6-v2)
- Synthesizer LLM: Google Gemini 2.5 Flash-Lite
- Verifier LLM: Llama 3.3 70B via Groq (different vendor from synthesizer)
- Data source: PubMed E-utilities API
- Optional desktop demo frontend: Flutter (frontEndapp/)
- Deployment:
 - Frontend: Vercel
 - Backend: Hugging Face Spaces (Docker)

2) Deployed Project Links (No local build required)

If you only want to evaluate functionality, use the hosted version:

- App (Frontend): <https://abhi04-medcite.vercel.app>
- Backend API: <https://Tony0489-MedCite-api.hf.space>
- Health check: <https://Tony0489-MedCite-api.hf.space/health>

Note: You do not need to rebuild ingestion data to use the deployed app.

3) Run Locally (Optional)

Prerequisites

- Python 3.12+
- Node.js 22 LTS + npm
- (Optional) Flutter SDK for Windows desktop demo

Environment Variables (Required for local backend)

Create a root .env file at:

- e:\Development\jubilant hackathon\.env

You can copy from template:

```
cd "e:\Development\jubilant hackathon"  
Copy-Item .env.example .env
```

Add these values in .env :

```
GOOGLE_API_KEY=your_google_ai_studio_key  
GROQ_API_KEY=your_groq_key  
NCBI_API_KEY=your_ncbi_key_optional  
NCBI_EMAIL=your_email@example.com
```

Where to get keys:

- Google Gemini key (`GOOGLE_API_KEY`): <https://aistudio.google.com/app/apikey>
- Groq key (`GROQ_API_KEY`): <https://console.groq.com/keys>
- NCBI key (`NCBI_API_KEY` , optional but recommended): <https://www.ncbi.nlm.nih.gov/account/settings/>

Notes:

- `NCBI_API_KEY` is optional; without it, PubMed still works but with lower rate limits.
- Never commit `.env` or `frontend/.env.local` to git.

Backend (Terminal 1, PowerShell)

```
cd "e:\Development\jubilant hackathon\backend"
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
uvicorn main:app --host 0.0.0.0 --port 8000 --reload
```

Backend (Terminal 1, Git Bash)

```
cd "/e/Development/jubilant hackathon/backend"
python -m venv .venv
./.venv/Scripts/activate
pip install -r requirements.txt
uvicorn main:app --host 0.0.0.0 --port 8000 --reload
```

Frontend (Terminal 2)

```
cd "e:\Development\jubilant hackathon\frontend"
npm install
```

Frontend (Terminal 2, Git Bash)

```
cd "/e/Development/jubilant hackathon/frontend"
npm install
npm run dev
```

Create `frontend/.env.local` with:

```
NEXT_PUBLIC_API_URL=http://localhost:8000
```

Then run:

```
npm run dev
```

Verify

- Frontend: `http://localhost:3000`
- Backend health: `http://localhost:8000/health`

4) Demo Prompts

Use the curated prompt set here:

- [DEMO_PROMPTS.md](#)

Hero query set (judge demo order) is documented in:

- [PITCH.md](#)

5) Common Issues and Fix Commands

Issue A: "Backend offline" in UI

Cause: backend not running, wrong API URL, or cold backend startup.

Fix:

```
# confirm backend health
Invoke-RestMethod -Method GET -Uri http://localhost:8000/health
```

```
# confirm backend health
curl -s http://localhost:8000/health
```

Also verify `frontend/.env.local` :

```
NEXT_PUBLIC_API_URL=http://localhost:8000
```

Then restart frontend:

```
cd "e:\Development\jubilant hackathon\frontend"
npm run dev
```

Issue B: Missing API key error (`GOOGLE_API_KEY` / `GROQ_API_KEY`)

Cause: `.env` missing backend secrets.

Fix:

1. Copy `.env.example` to `.env` at repo root.
2. Fill required keys.
3. Restart backend server.

Issue C: LanceDB table not found

Cause: local vector index not initialized.

Fix:

```
cd "e:\Development\jubilant hackathon\backend"
.\.venv\Scripts\Activate.ps1
python -m ingestion.embedder
```

```
cd "/e/Development/jubilant hackathon/backend"
./.venv/Scripts/activate
python -m ingestion.embedder
```

Issue D: PowerShell blocks `sync.ps1` ("running scripts is disabled")

Cause: Execution policy restriction.

Fix (current session only):

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

Then run your sync command again.

Issue E: Port already in use (`8000` or `3000`)

Fix (example using alternate ports):

```
# backend
uvicorn main:app --host 0.0.0.0 --port 8001 --reload
```

```
# backend
uvicorn main:app --host 0.0.0.0 --port 8001 --reload
```

```
# frontend/.env.local
NEXT_PUBLIC_API_URL=http://localhost:8001
```

Then restart frontend.

6) Optional Flutter Desktop Demo

```
cd "e:\Development\jubilant hackathon\frontEndapp"
flutter pub get
flutter run -d windows --dart-define=MEDCITE_BASE_URL=https://Tony0489-MedCite-api.hf.space
```

7) Repository

- GitHub: <https://github.com/Abhishek0489/medCite>